MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. $(\sec A + \tan A)(1 - \sin A)$ is equal to: [1]

(A) $\sec A$ (B) $\sin A$ (C) $\csc A$ (D) $\cos A$

Q2. $\frac{1+\tan^2 A}{1+\cot^2 A}$ is equal to: [1]

(A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$

Q3. If $\cos\theta + \cos^2\theta = 1$, then the value of $\sin^2\theta + \sin^4\theta$ is: [1]

(A) -1 (B) 0 (C) 1 (D) 2

Q4. If $\sec\theta + \tan\theta = x$, then the value of $\sec\theta - \tan\theta$ is: [1]

(A) x^2 (B) $\frac{1}{x}$ (C) $-x$ (D) $1 - x$

Q5. Using the previous result, if $\sec\theta + \tan\theta = x$, then $\sec\theta$ is equal to: [1]

(A) $\frac{x^2+1}{2x}$ (B) $\frac{x^2-1}{2x}$
(C) $\frac{x^2+1}{x}$ (D) $\frac{2x}{x^2+1}$

Q6. **Assertion (A):** The value of $\frac{1-\cos^2\theta}{\sin^2\theta}$ is equal to 1 for all acute angles θ.
Reason (R): $\sin^2\theta + \cos^2\theta = 1 \implies 1 - \cos^2\theta = \sin^2\theta$, and for acute angles θ, $\sin\theta \neq 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. In ΔPQR, right-angled at Q, $PQ = 3$ cm and $PR = 6$ cm. Determine $\angle QPR$ and $\angle PRQ$. [2]

Q8. If $\cos\theta = \frac{1}{2}$, find the value of $\frac{2\sec\theta}{1+\tan^2\theta}$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Prove that: $\frac{\cos^3\theta+\sin^3\theta}{\cos\theta+\sin\theta} + \frac{\cos^3\theta-\sin^3\theta}{\cos\theta-\sin\theta} = 2.$ [3]

Q10. Prove that: $\frac{\sin A}{1-\cos A} = \csc A + \cot A.$ [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

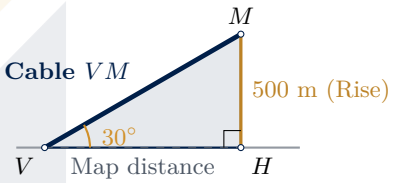
Q11. Prove that: $\frac{\cos A-\sin A+1}{\cos A+\sin A-1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$. [5]

Q12. If $x = a\cos^3\theta$ and $y = b\sin^3\theta$, prove that $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Mountain Cable Car / Aerial Tramway Span

A passenger ropeway connects a valley base station V to an alpine mountain peak station M. The cable runs straight between the two stations along an incline angle of $\theta = 30°$ to the horizontal. The vertical elevation rise between the two stations is $h = 500$ m.



Based on the above geometric model, answer the following questions:

(i) Find the total direct track length VM of the ropeway cable. [1]

(ii) Find the horizontal map distance between the base station and mountain station. [1]

(iii) If a cable car travels at a constant speed of 5 m/s, calculate the travel time in seconds and minutes from the valley station to the summit. [2]

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